FORM 2 - TERMS OF REFERENCE (TOR)

Project Title : ***Mangrove Ecosystem Services Assessment for Conservation Strategies and Stakeholders' Willingness to Pay in San Miguel Bay***

Principal Investigator : For. Divina P. Chica
Implementing Institution (HEI) : Partido State University-Tinambac Campus

Collaborating Institution/s : Local Government Units / Municipalities of:
(1) Mercedes, Camarines Norte
(2) Basud, Camarines Norte
(3) Sipocot, Camarines Sur
(4) Cabusao, Camarines Sur
(5) Calabanga, Camarines Sur
(6) Tinambac, Camarines Sur
(7) Siruma, Camarines Sur

Period Covered : Two (2) years
Total Project Cost : Php 4,500,000.00

Objectives

General Objective: To determine the economic value of ecosystems services offered by the existing mangrove forest vegetation in San Miguel Bay

Specific Objectives:

1. Analyze the community structure, density and diversity of mangrove forest along San Miguel Bay.
2. Assess the ecosystem services of mangrove forest and the socio-psychological factors that support the sustainable productivity
3. Analyze the fisheries of mangrove ecosystem.
4. Determine the willingness-to-pay (WTP) of mangrove-associated communities along SMB.
5. Propose strategies and tools towards community-based mangrove conservation and management.

Program/Project Scope:
This research study covers the localities surrounding the coastal areas of San Miguel Bay (SMB), namely: municipalities of Mercedes and Basud in Camarines Norte, and municipalities of Sipocot, Cabusao, Calabanga, Tinambac, and Siruma in Camarines Sur. In addition, the coverage of the research study is limited to the indirect use value of mangrove forests of SMB, in relation to fishery and coastal protection.

Project Methods (Sampling, Design and Data Treatment):

1. Community Structure and Biodiversity Assessment of Mangrove Ecosystem in San Miguel Bay

Sampling
The conduct of vegetation analysis and carbon stock assessment of Mangrove Forests along San Miguel Bay will follow the standard technique, wherein the sampling frame includes all the areas (marked as sampling sites) illustrated in Figure 5.

For the determination of community structure, the sampling design that will be employed is stratified quadrat sampling technique that will apply three stratifications, i.e location (municipality), topographic position (seaward, midward, landward), and mangrove forest type (scrub, fringing, hammock, riverine, basin, overwash). This is the same technique that was implemented by Malabrigo et al. (2017) in the mangrove forest of Pagbilao, Quezon. The belt transect method will be applied to ensure that all zones of the forest are sampled and will follow a distance of 100-250 meters between transects, depending on the length of the sampling area. 10m x 10m plots shall also be established along the transect lines with intervals of 50 or 25 meters, depending on the extent of the stands.

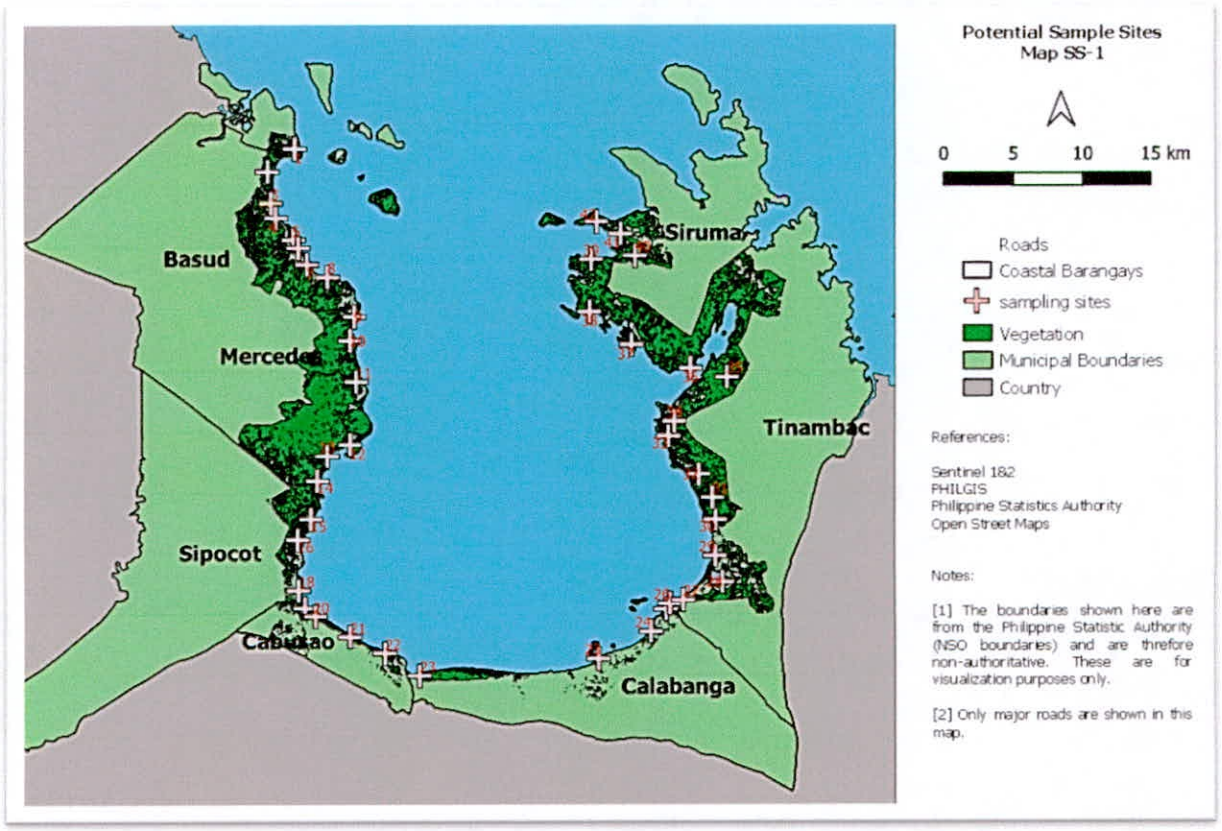


Figure 5. Identified sampling frame of Mangrove Forest along San Miguel Bay.

Data Collection

Identification of all species within the sampling quadrats will be made. Collection of samples will be made in case of unidentified mangrove species.

For the community structure, within the 10 by 10-meter plot, the following parameters will be measured: diameter at breast height (DBH) and total height (TH) of all individual species with ≥ 5 cm DBH. Tree diameter measurement will be at 1.3 m aboveground or 10 cm above the tallest buttress/prop roots if taller than 1.3 m. In mangrove trees with tree forks below 1.3 m from the ground, all stems with ≥ 5 cm DBH will be measured separately.

Data Analysis

All information will be encoded and organized.

For the community structure, the following will be determined: relative density, relative dominance, and relative frequency values of each tree species. This set of information will be used to determine the importance value (IV) of each species. IVs will be computed using the following formula:

Density = number of individuals / areas sampled

Relative Density = (density for a species / total density for all species) x 100

Frequency = number of plots in which species occur / total number of plots sampled

Relative Frequency = (frequency value for a species /total frequency for all species) x 100

Dominance = basal area or volume for a species/ area sampled

Relative Dominance = (dominance for a species/ total dominance for all species) x100

Importance Value = Relative Density + Relative Frequency + Relative Dominance

Diversity indices (Shannon and, Simpson's Evenness) for the collected data will also be computed using the following equations below. Shannon index is a measure of species abundance and richness to quantify diversity of the species within an area of study while Simpson's index (D) is a measure of proportional abundance and measures the

probability that two individuals randomly selected from a sample will belong to the same species (Moutsambote et. al., 2016).

$$\text{Shannon Index (H)} = -\sum_{i=1}^{s} p_i \ln p_i \qquad \text{Simpson Index (D)} = \frac{1}{\sum_{i=1}^{s} p_i^2}$$

Where:

p = proportion of individuals (n/N)
n = number of one particular species found (n)
N = total number of individuals found
ln = the natural log,
Σ = the sum of the calculations,
s = the number of species.

2. Inventory of Ecosystem Services of Mangrove Forests along San Miguel Bay

Sampling
Respondents will be surveyed on the absence and presence of the ecosystem services provided in questionnaires; The sampling framework is the total number of households in all the municipalities surrounding the San Miguel Bay. The sampling design that will be applied to the sampling framework is simple random sampling. The sample was also obtained using the Slovin's formula, which is computed as follows: n = N / (1+Ne2).
Where: n = sample size; N= population size; and e=margin of error

The total number of households (N) in the study site is 73,771. In addition, the margin of error (e) that will be assumed for this study is 5%. Given this set of information, the sample size was computed as follows:

Sample size = Total number of Households / (1+Total number of Householdsx0.05^2)

Sample Size = 73, 771 / (1+73,771 x 0.05^2) ➔ 397.85 or 398 households

Data Collection
A survey will be conducted utilizing the standardized questionnaire and specifically assess for the presence and absence of the services. To validate the information gathered thru stakeholders' survey, a triangulation of data thru key informant interview and focused-group discussion will also be conducted. The standardized list of ecosystem services which will be used to assess the presence and absence of ES, based from the preferences of the stakeholders, will be adopted from the version of the ecosystem services framework proposed by the MA (2003) and CICES (2013).

Data Analysis
All information that will be collected will be encoded accordingly. Since identification and quantification of mangrove forests ecosystem services will be considered in this study, descriptive analysis of the data collected will be utilized.

3. *Assessment of Fisheries Production of Mangrove Ecosystem in San Miguel Bay.*

Sampling and Data Collection

Macro- Invertebrate

The study will employ Rapid Resource Assessment, making use of survey questionnaires given to identify fisher folks, particularly gleaners, as main data collection tools, the same method used by Nieves et al. (2010). Other information such as catch volumes, catch rates, income, and macro-invertebrates' species can be assessed through the survey forms and during key informants' interviews and focus group discussions on the coastal barangays of the seven municipalities.

Fishing Gear and Catch per Unit Effort (CPUE)
CPUE is a method used to estimate fishery resource stocks based on the used fishing gear. The method of sampling for fishing gears will be through surveys and key informants' interviews (KII) using stratified random sampling. Additional available data will be collected from the region's office of the Department of Agriculture. Information such as time of sampling and weather condition will also be noted. Weather conditions will be retrieved from PAG-ASA to monitor possible influence of season on species caught.

Data Analysis

CPUE analysis: Descriptive analysis will be used in describing the average catch per unit effort, fishing gears used and type of species caught. ANOVA will compare the result from the gears used. Other data analysis such as Regression, and Pearson's Correlation Coefficient will be calculated using statistical tools such as FAST or R.

The Catch Per Unit Effort will be calculated using Gunarso & Wiyono, (1994) formula:

$CPUE = \frac{Ci}{Ei}$ or $CPUE = \frac{Total\ Weight\ of\ Catch\ (kg)}{Effort\ (trips, hours, no.of\ man)}$

If a resource/species is caught by different fishing gears, it is important to standardize the use of fishing gears because each gear has different use and perception (Jatmiko et al 2014). The standardization is calculated by finding the value of power indices of capture (Fishing Power Index/FPI) of each gear (Gunarso & Wiyono, 1994) is:

$$FPIi = \frac{CPUEi}{CPUEs}$$

Where:
FPIi is fishing power index of specific fishing gear
CPUEi is catch per unit effort from fishing gear to be standardized
CPUEs is catch per unit effort from standardized fishing gear

And continue to:

$$fs = \sum_{i=1}^{n} FPIi\ x\ fi$$

Where:
fs is standardized effort (unit); and
fi is effort (fishing gear) to be standardized (unit).

To calculate the income per unit effort:
IPUE= CPUE (kg/hr-man) x buying price (PhP/kg)

To determine diversity of macro-invertebrates' species, Shannon-diversity index will be utilized.

Shannon-diversity index: $$H' = -\sum\left[\left(\frac{n_i}{N}\right)\times\ln\left(\frac{n_i}{N}\right)\right]$$

Where:
ni = number of individuals or amount (e.g., biomass) of each species (the ith species)
N = total number of individuals (or amount) for the site
ln = the natural log of the number.

Species richness will be calculated using Menhinick's index which is simply the number of species (*n*) divided by the square-root of the total number of individuals (*N*).

$$D = nN\text{--}\sqrt{}$$

Correlation analysis is used to determine the closeness of the relationship between two variables, namely the correlation that occurs between fish production and mangrove forest area, the formula for the correlation test is:
iy =Capture fisheries production (kg)
i x =area of mangrove forest (hectares)
R=Correlation coefficient
i=Year

There are two stages in determining the relation of mangroves to fishery production. The first stage is the technical efficiency which is the stochastic frontier production which is:

$$\ln(Q_{it}) = \beta_0 + \beta_1 \ln(x_{1it}) + \beta_2 \ln(x_{2it}) + v_{it} - u_{it}$$

Where:
bis = the parameters to be estimated
x1 = fisheries outlay
x2 = fishing vessels
Q is marine fish output
I = coastal state
t = year

In the second stage of the analysis, the influence of mangroves on technical efficiency will be ascertained. Since the estimated technical efficiency values are bound between 0 and 1, they are normalized before the regression analysis is undertaken. The specification of the second-stage panel (fixed effects) regression model is:

$$\ln[TE_{it}/(1-TE_{it})] = \alpha_0 + \alpha_1 MF_{it} + \sum_{i=1}^{n-1} \beta_i SD_i + e_{it}$$

Where: TE is technical efficiency,
MF is mangrove fringe (defined as the square root of mangrove area)
SD represents the dummy variables that control for unobserved state fixed

4. *Contingent Valuation Study on Mangrove Forest Ecosystem Conservation and the fisher folks and coastal communities in San Miguel Bay*

Sampling
The sampling framework of this study is the total number of households in all the municipalities surrounding the San Miguel Bay. The sampling design that will be applied to the sampling framework is stratified random sampling, using two stratifications, namely a) location (municipality) and b) level of engagement or exposure to mangrove forest, based on the distance of their residence to the coastal areas/ mangrove communities (high: seaward, medium: midward, and low: landward) to proportionately represent the communities in the respective municipalities.

The sample was also obtained using the Slovin's formula, which is computed as follows: n = N / (1+Ne2).
Where: n = sample size; N= population size; and e=margin of error

The total number of households (N) in the study site is 73,771. In addition, the margin of error (e) that will be assumed for this study is 5%. Given this set of information, the sample size was computed as follows:

Sample size = Total number of Households / (1+Total number of Householdsx0.05^2)

Sample Size = 73, 771 / (1+73,771 x 0.05^2) ➔ 397.85 or 398 households

Data Collection

As mentioned above, the data collection will be stratified according to location and exposure or engagement of the households to mangrove communities.

The following are mixed methods of data collection adopted from a CVM study of Tuan et. al. in 2014, Secondary data will be collected from the involved local government units to gather socio-economic and demographic data to be able to identify various stakeholders for the study. Key informant interviews will be done to identify vital issues in relation to the utilization, management and conservation of mangrove resources along the coast of SMB. Focus group discussions will be the core method in collecting qualitative information and will include the identified stakeholders to induce information exchange to provide initial ideas that will then be developed into different CV scenarios. Lastly, an initial household questionnaire survey will be designed and pre-tested to guarantee and/or further improve the final CV questionnaire.

Data Analysis
All information that will be collected will be encoded accordingly. The socio-economic profile and the CVM answers will be organized according to standard layout for STATA. The data in CVM will be subjected to linear regression analysis, specifically probit analysis.

5. *Socio-Psychological Factors affecting Utilization and Willingness to participate in the Management and Conservation of Mangrove Ecosystem of San Miguel Bay*

The following are adopting the approaches of Abdullah et. al. (2016) for the psychological factors affecting mangrove utilization and Zhang et. al. (2011) for the factors influencing willingness to participate in the management and conservation of mangrove ecosystems of SMB.

Sampling

The sampling method and sample size for the household survey will be the same with the fourth (4th) component of study. However, there will be a qualitative data collection in terms of KIIs and FGD where a purposive sampling method shall be implemented for the key stakeholder of the mangrove ecosystem of SMB.

Data Collection

The data collection for this component is a sequential mixed-method strategies and will be combining quantitative and qualitative data collection. There will be a total of 398 households to be interviewed, preferably the head of the family but if none, any member of the family aged 18 years old and older shall be interviewed. The questionnaire to be used will have the following as components but not necessarily limited to:

- Letter of consent
- Socio-demographic information and household characteristics section
- Enumeration of practices and level of mangrove utilization
- Knowledge, beliefs, attitude towards mangrove utilization
- Level of perception and willingness to participate in the management and conservation of Mangrove ecosystem of SMB

FGD and KIIs will be used to collect qualitative data from key stakeholders and will include semi-structured questions based from triangulation of data gathered from the household survey.

Data Analysis

Gathered information will be collected and encoded accordingly. Quantitative descriptive analysis shall be employed to present the socio-economic data, mangrove utilization practices, factors and willingness to participate in percentages. Pearson chi-square test will be used to determine the correlation between WTP and socio-demographical information of the respondents. Collected data from FGD and KIIs will be used to further the discussion of the results of the survey.

Activities (Year 1)	Months											
	1	2	3	4	5	6	7	8	9	10	11	12
Component 1: Community Structure and Biodiversity Assessment of Mangrove Ecosystem in San Miguel Bay												
1.Pre-implementation activities (meetings with LGUs)	■	■	■									
2. Formulation of MOA/MOU with LGUs and other concerned institutions	■	■	■									
3. Secondary Data Collection		■	■	■	■	■	■	■	■			
4. Initial Site visit and identification of sampling plots			■	■	■							
5. Primary Data Collection (Field work)				■	■	■	■	■	■			
6. Project Assessment Meeting, Evaluation and Monitoring with CHED and ParSU Project Team											■	
7. Submission of Inception Report				■								
8. Submission of Progress Reports (Quarterly)			■			■			■			■
9. Reviewing and cleaning of gathered data								■	■			
10. Data Processing and Analysis									■	■	■	
11. Final/ Terminal Report and Output Preparation and Writing										■	■	■
12. Submission of final manuscript												■
13. Writing of Publishable Articles in Reputable Journals (ISI-Scopus, ACI Indexed or CHED JIP Journals)												■
Component 2: Inventory of Ecosystem Services of Mangrove Forests along San Miguel Bay												
1.Pre-implementation activities (IEC meetings with LGUs)	■	■	■									
2. Formulation of MOA/MOU with LGUs and other concerned institutions	■	■	■									
3. Secondary Data Collection		■	■	■	■	■	■	■	■	■	■	■
4. Primary Data Collection (FGD & KII)			■	■	■	■	■	■	■	■	■	■
5. Data analysis and Discussions			■	■				■	■			■
6. Project Assessment Meeting, Evaluation and Monitoring with CHED and ParSU Project Team											■	
7. Submission of Inception Report				■								
8. Submission of Progress Reports (Quarterly)			■			■			■			■

Activities (Year 1)	Months											
	1	2	3	4	5	6	7	8	9	10	11	12
Component 3a: Assessment of Macro-invertebrates and Catch Per Unit Effort of Fishing Gears for Fishery Profile of Coastal Communities in San Miguel Bay												
1.Pre-implementation activities (IEC meetings with LGUs)	■	■	■									
2. Formulation of MOA/MOU with LGUs and other concerned institutions	■	■	■									
3. Secondary Data Collection		■	■	■	■	■	■	■	■	■	■	■
4. Primary Data Collection (FGD & KII)			■	■	■	■	■	■	■	■	■	■
5. Project Assessment Meeting, Evaluation and Monitoring with CHED and ParSU Project Team											■	
6. Submission of Inception Report				■								
7. Submission of Progress Reports (Quarterly)			■			■			■			■
Component 4: Contingent Valuation Study on Mangrove Forest Ecosystem Conservation and the fisher folks and coastal communities in San Miguel Bay												
1.Pre-implementation activities (IEC meetings with LGUs)	■	■	■									
2. Formulation of MOA/MOU with LGUs and other concerned institutions	■	■	■									
3. Secondary Data Collection		■	■	■	■	■	■	■	■	■	■	■
4. Primary Data Collection (FGD & KII)			■	■	■	■	■	■	■	■	■	■
5. Primary Data Collection (Household Survey)							■	■	■	■	■	■
6. Project Assessment Meeting, Evaluation and Monitoring with CHED and ParSU Project Team											■	
7. Submission of Inception Report				■								
8. Submission of Progress Reports (Quarterly)						■			■			■
Component 5: Socio-Psychological Factors affecting Utilization and Willingness to Participate in the Management and Conservation of Mangrove Ecosystem of San Miguel Bay												
1.Pre-implementation activities (IEC meetings with LGUs)	■	■	■									
2. Formulation of MOA/MOU with LGUs and other concerned institutions	■	■	■									
3. Secondary Data Collection		■	■	■	■	■	■	■	■	■	■	■
4. Primary Data Collection (Household Survey)			■	■	■	■	■	■	■	■	■	■
5. Triangulation of gathered data through FGDs and KIIs							■	■	■	■	■	■
6. Project Assessment Meeting, Evaluation and Monitoring with CHED and ParSU Project Team											■	
7. Submission of Inception Report				■								
8. Submission of Progress Reports (Quarterly)			■			■			■			■
9. Data encoding and treatment						■	■	■				
10. Data analysis and Discussions							■	■	■			

Activities (Year 2)	Months											
	13	14	15	16	17	18	19	20	21	22	23	24
Component 1: Community Structure and Biodiversity Assessment of Mangrove Ecosystem in San Miguel Bay												
1. Writing of Publishable Articles in Reputable Journals (ISI-Scopus, ACI Indexed or CHED JIP Journals)	■											
2. Submission of Progress Reports (Quarterly)			■			■			■			
3. Creation of IEC materials from gathered data	■	■	■									
4. Crafting of manual on the mangrove species of SMB	■	■	■	■	■							
5. Approval of Crafted Manual					■	■	■					
Component 3a: Assessment of Macro-invertebrates and Catch Per Unit Effort of Fishing Gears for Fishery Profile of Coastal Communities in San Miguel Bay												
1. Secondary Data Collection	■	■										
2. Primary Data Collection (FGD & KII)	■	■										
3. Primary Data Collection (Household Survey)			■			■			■			■
4. Data Management and Analysis			■	■	■					■	■	
5. Submission of Progress Reports (Quarterly)					■						■	■
6. Final/ Terminal Report and Output Preparation and Writing											■	■
7. Submission of Draft Terminal Report			■	■	■					■	■	
8. Submission of Final Report					■	■					■	■
9. Writing of Publishable Articles in Reputable Journals (ISI-Scopus, ACI-Indexed, CHED JIP)											■	■
10. Reporting of Results to LGUs and other partner institutions and communities								■				■
Component 3b: Assessment of Fisheries Production of Mangrove Ecosystem in San Miguel Bay												
1. Secondary Data Collection	■	■	■	■	■	■	■	■	■			
2. Primary Data Collection (FGD & KII)	■	■	■	■	■	■	■	■	■			
3. Primary Data Collection (Household Survey)	■	■	■	■	■	■	■	■	■			
4. Data Management and Analysis	■	■	■	■	■	■	■	■	■			
5. Submission of Progress Reports (Quarterly)			■			■			■			
6. Submission of Draft Terminal Report											■	
7. Final/ Terminal Report and Output Preparation and Writing											■	■
8. Writing of Publishable Articles in Reputable Journals (ISI-Scopus, ACI Indexed, CHED JIP)											■	■
9. Reporting of Results to LGUs and other partner institutions and communities												■
Component 4: Economic Valuation of Mangrove Ecosystem in San Miguel Bay												
1. Secondary Data Collection	■	■	■	■	■	■						
2. Primary Data Collection (FGD & KII)	■	■	■	■	■	■						
3. Primary Data Collection (Household Survey)	■	■	■	■	■	■						
4. Data Management and Analysis	■	■	■	■	■	■						
5. Submission of Progress Report (Quarterly)			■			■			■			
Final/ Wrap-up Project Activities												
Project Assessment Meeting, Evaluation, Terminal Monitoring with CHED and ParSU Project Team										■		
Final/ Terminal Report and Output Preparation and Writing							■	■	■	■		
Preparation of Draft Terminal Report										■		
Submission of Final Report										■	■	
Writing of Publishable Articles in Reputable Journals (ISI-Scopus, ACI Indexed or CHED JIP Journals)										■	■	■
Reporting of Results to LGUs and other partner institutions and communities												■

Personnel Requirements/Team Composition & Modus Operandi (including delineation of assignments, coordination/networking arrangements):

Program/Project Activities	Person Involved/ Institutional Affiliation	Responsibility / Scope of Work	Duration of Involvement/ Time Frame (in days/months)
Estimating mangrove forest cover using remote sensing	Carlo Gabriel Abante/ PSU	1. Delineate the area of research 2. Generate shapefiles required in the estimation of mangrove forest cover 3. Encoding of attributes, interpreting and translating visual representation to the actual attributes of the area	1. Month 1 to Month 24 (5 days per month) 2. Month 1 to Month 24 (5 days per month) 3.Month 1 to Month 24 (5 days per month)
Secondary data collection	Carlo Gabriel Abante / PSU Ceazar Ian Pesimo / PSU Mikka Ella Cabrera / PSU Divina Chica / PSU Engr. Irene Asetre/ PSU Project Staff	1. Gather secondary data and related literatures concerning: a. Vegetation analysis using GIS and remote sensing b. Common drivers affecting human behavior towards the use of natural resources c. Assessment Fishery Production in Mangrove Ecosystem d. WTP and CVM methods and practices	Month 1 to Month 24 (5 days per month) Month 1 to Month 24 (5 days per month) Month 1 to Month 24 (5 days per month) Month 1 to Month 24 (5 days per month)
Community Structure and Biodiversity Assessment of Mangrove Ecosystem in San Miguel Bay	Ceazar Ian Pesimo / PSU Divina Chica / PSU Carlo Gabriel Abante/ PSU Mikka Ella Cabrea/ PSU Engr. Irene Asetre/ PSU Project Staff	1. Determine the sampling size 2. Establish the inventory plots 3. Conduct tree inventory 4. Data Management 5. Manuscript write-up 6. Write-up of other deliverables (IEC and Manual)	1. Month 1 to Month 12 (5 days per month) 2. Month 1 to Month 12 (5 days per month) 3.Month 1 to Month 12 (10 days per month) 4. Month 1 to Month 12 (5 days per month) 5. Month 1 to Month 12 (5 days per month) 6. Month 13 to 24 (5 days per month)
Socio-Psychological Factors affecting Utilization and Willingness to Participate in the Management and Conservation; and Inventory of Ecosystem Services of mangroves stands in San Miguel Bay	Engr. Irene Asetre/ PSU Ceazar Ian Pesimo / PSU Dr. Raul G. Bradecina/ PSU Divina Chica / PSU Carlo Gabriel Abante/ PSU Mikka Ella Cabrea/ PSU Project Staff	1. Determine the sampling population 2. Questionnaire development 3. Conduct household surveys, KIIs and FGDs 4. Data Management 5. Manuscript write-up. 6. Write-up of other deliverables (IEC and Manual)	1. Month 1 to Month 12 (5 days per month) 2. Month 1 to Month 12 (5 days per month) 3.Month 1 to Month 12 (10 days per month) 4. Month 1 to Month 12 (5 days per month) 5. Month 1 to Month 12 (5 days per month) 6. Month 13 to 24 (5 days per month)
Assessment of Fisheries Production of Mangrove Ecosystem; and Assessment of Macro-invertebrates and Catch Per Unit Effort of Fishing Gears for Fishery Profile of Coastal Communities in San Miguel Bay	Mikka Ella Cabrea/ PSU Dr. Raul G. Bradecina/ PSU Divina Chica / PSU Ceazar Ian Pesimo / PSU Carlo Gabriel Abante/ PSU Engr. Irene Asetre/ PSU Project Staff	1. Determine the sampling population 2. Questionnaire development 3. Conduct household surveys, KIIs and FGDs 4. Data Management 5. Manuscript write-up. 6. Write-up of other deliverables (IEC and Manual)	1. Month 1 to Month 12 (5 days per month) 2. Month 1 to Month 12 (5 days per month) 3.Month 1 to Month 24 (10 days per month) 4. Month 1 to Month 24 (5 days per month) 5. Month 1 to Month 24 (5 days per month) 6. Month 13 to 24 (5 days per month)
Contingent Valuation Study on Mangrove Forest Ecosystem Conservation and the fisher folks and	Divina Chica / PSU Dr. Raul G. Bradecina/ PSU Ceazar Ian Pesimo / PSU Carlo Gabriel Abante/ PSU	1. Determine the sampling population 2. Questionnaire development and series of pretesting	1. Month 1 to Month 12 (5 days per month) 2. Month 1 to Month 12 (5 days per month)

coastal communities in San Miguel Bay	Engr. Irene Asetre/ PSU Mikka Ella Cabrea/ PSU Project Staff	3. Conduct household surveys, KIIs and FGDs 4. Data Management 5. Manuscript write-up. 6. Write-up of other deliverables (IEC and Manual)	3.Month 1 to Month 18 (10 days per month) 4. Month 1 to Month 18 (5 days per month) 5. Month 1 to Month 18 (5 days per month) 6. Month 13 to 24 (5 days per month)
Progress Report Write-up	Carlo Gabriel Abante / PSU Ceazar Ian Pesimo / PSU Divina Chica / PSU Mikka Ella Cabrera / PSU Engr. Irene Asetre / PSU Dr. Raul G. Bradecina/ PSU	1. Integrate and prepare progress reports on: a. Mapping and vegetation assessment b. Ecosystem services inventory c. Stakeholders' perception, behavior and socio-psychological factors towards utilization and management of mangrove forests d. Assessment of Fishery Productivity and Macro-invertebrates and catch per unit effort of fishing gears e. Economic Value of the Productivity of the Mangrove in San Miguel Bay	a. Month 1 to Month 24 (5 days per month) b. Month 1 to Month 24 (5 days per month) c. Month 1 to Month 24 (5 days per month) d. Month 1 to Month 24 (5 days per month) e. Month 1 to Month 24 (5 days per month)
Final Report Write-up	Carlo Gabriel Abante / PSU Ceazar Ian Pesimo / PSU Divina Chica / PSU Irene Astre / PSU Mikka Ella Cabrera / PSU Dr. Raul G. Bradecina/ PSU	1. Data integration and final report writeshop 2. Write-up for the IEC and Manual	Month 1 to Month 24 (10 days per month)

Personnel under Contract of Service

Person Involved/ Institutional Affiliation	Duties/Responsibilities and Deliverables	Duration of Involvement/ Time Frame
Project Support Staff IV One (1) pax - Must be a baccalaureate degree holder - Field specialization: Environmental Science and Environmental Management - Must be organized, and proficient in oral and written skills. - Must be skillful in basic photoshop and editing.	1. Provide assistance in the conduct of activities under the research project by: - Coordinating with the identified collaborating institutions to facilitate meetings, conduct of KIIs, FGDs, and Household Surveys for components 2, 3, 4, and 5. - Preparing communication letters with different institutions - Handling logistics and other necessary preparations in line with the conduct of research activities of all components - Being in-charge with encoding data collected during fieldworks and interviews 2. Perform basic photoshop, editing and layouting for the production of manual: Mangroves of SMB. 3. Perform other duties and responsibilities in line with the activities of the research study.	Month 1 to Month 24
Project Support Staff V One (1) pax - Must be a baccalaureate degree holder - On-going or MS degree holder is an advantage - Field specialization: Environmental Science, Forestry, Biology and other related fields. - Must be organized, and proficient in data analyzing skills. - Must be knowledgeable on mangrove inventory.	1. Provide assistance in the conduct of activities under the research project by: - Participating in field data gathering activity of component 1. - Participating in the conduct of KIIs, FGDs, and Household Surveys for components 2, 3, 4, and 5. - Being of aid in organizing data collected during fieldworks and interviews - Photo-documenting the research activities & capturing high quality images to be used in the Manual. 2. Support the editing and layouting for the production of manual: Mangroves of SMB. 3. Perform other duties and responsibilities in line with the activities of the research study.	Month 1 to Month 24

SUBMITTED BY:

For. DIVINA P. CHICA
Principal Investigator (Project Leader)
Partido State University-Tinambac Campus

REVIEWED AND ENDORSED BY:

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